

Retrosynthesis - Reaction Center Classification

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Class: Deep Learning

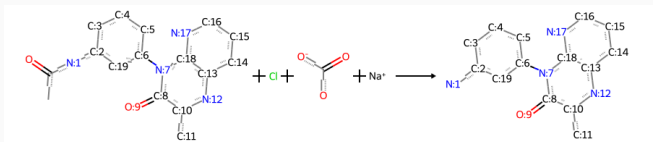
Professor: G. Mpouritsas

Introduction to the Problem

What is Retrosynthesis?

The process of breaking down a target molecule into simpler, available precursors.

- **The Challenge:** It's a complex one-to-many problem. A single product can be made in many ways.
- **Our Focus: Reaction Center Prediction.** This is the critical first step: identifying the exact bond(s) in the product that were formed during the reaction.



Common Methodologies

Approaches for this task generally fall into three categories:

- **Template-Based:** Matches products to a fixed library of reaction rules (e.g., SMARTS patterns). While interpretable, they struggle to generalize to new, unseen reactions.
- **Template-Free:** Uses sequence-to-sequence models to "translate" a product into reactants. This is more flexible but less interpretable and can produce invalid molecules.
- **Semi-Template:** A hybrid approach that splits the task into two stages. Predicting the reaction center, then constructing the reactant (usually utilising some template knowledge at this step)

Our Approach

We treat molecules as graphs and use a GNN to learn from their structure.

Data and Representation

- **Data:** We use the USPTO-50k dataset, a standard benchmark for reaction prediction.
- **Graph Representation:** We use RDKit and PyTorch Geometric to convert chemical structures into graphs where atoms are nodes and bonds are edges.
 - It's a **natural representation** of molecular structure.
 - Avoids grammatical headaches of SMILES
 - Easier learning from an atom's **local and global chemical environment**.

Multi-Task Graph Neural Network

Multi-Task Learning

Our GNN learns to solve two problems at once:

- **Bond Prediction:** How likely is it that a bond will break?
 - The central task of our approach.
- **Count Prediction:** For a given graph, how many bonds will break in total (0, 1, or 2)?
 - Eventually converted to probabilities and used as priors to enhance the Reaction Center prediction process.

Model Architecture

Our model is composed of four key modules:

1. Graph Encoder (TransformerConv)

Learns context-rich feature representations for each atom utilizing message passing.

2. Global Feature Extraction

Creates a single embedding representing the entire molecule by pooling all atom embeddings.

3. Bond-Head

A 2-layer MLP that predicts the probability of each individual bond breaking.

4. Count-Head

A separate 2-layer MLP that uses the global molecule features to predict the total number of bonds to break.

Algorithm 1 Model Training Pseudocode

Require: Training dataset D_{train} , Validation dataset D_{val}

```
1: function TRAINMODEL( $D_{train}, D_{val}$ )
2:   ratio  $\leftarrow \frac{\text{negative\_samples in } D_{train}}{\text{positive\_samples in } D_{train}}$ 
3:   Initialize optimizer with model parameters
4:   Initialize learning rate scheduler
5:   for each training epoch do
6:     for each batch in  $D_{train}$  do
7:        $\text{pred}_{\text{bonds}}, \text{pred}_{\text{count}} \leftarrow \text{model}(\text{batch})$ 
8:       loss  $\leftarrow \text{AggregatedLoss}(\text{predictions}, \text{ground\_truth}, \text{ratio})$ 
9:       Update the model's parameters using the calculated loss
10:    end for
11:    validation_metric  $\leftarrow \text{Evaluate}(\text{model}, D_{val})$ 
12:    Update learning rate with scheduler based on validation_metric
13:  end for
14:  return trained model
15: end function
```

Training - Metrics

AUPRC (Area Under the PR Curve)

- Our primary metric for validation and testing the bond-prediction task.
- Chosen to handle the massive class imbalance (positive class is only $\sim 2.86\%$).
- Measures the model's ability to find the rare positive instances (bonds that break).

Top-k Accuracy

- Evaluates performance on the test set by allowing up to k ranked predictions.
- A prediction is correct if the true answer is in the model's Top-1, 3, 5, or 10 suggestions.
- Simulates real-world use in retrosynthesis planners and is a standard metric for the field.

Count Prediction Accuracy

- Measures how often the model correctly predicts the exact number of bonds to break.
- Acts as a direct performance indicator for the model's Counting Head.

- **Data Flow:**

- USPTO-50k dataset processed into graph objects (atoms as nodes, bonds as edges) with engineered features.
- Split into 80% training, 10% validation, 10% test sets.
- Batches fed to the GNN, producing bond-level and molecule-level predictions.

- **Loss Function:**

- Aggregation of **Binary Cross Entropy** for bond prediction (handling class imbalance) and **Cross Entropy** for count prediction.
- $Loss = L_{bondBCE} + \lambda * L_{head_countCE}$

- **Optimizer & Scheduler:**

- **AdamW** optimizer for efficient parameter updates.
- **CosineAnnealingLR** scheduler, adjusting the learning rate for stable convergence.

Training - Hyperparameter Tuning

- Utilized **Optuna** for automated Bayesian optimization over 50 trials, each with 100 epochs.
- Early stopping after 5 consecutive epochs with no validation AUPRC improvement.
- **Starting Learning Rate:** 0.00099040204749014
- λ : 0.5665524341505498
- **Scheduler:** CosineAnnealingLR

Beam Search

Problem

Rigidly using the count-head's top guess (argmax) causes cascading failures. If the count-head underestimates breaks the prediction is guaranteed to fail.

The Solution: Beam Search

- Instead of a single guess, we generate a ranked list of the most likely **sets** of bonds (hypotheses).
- **Scoring:** Each hypothesis is scored by *combining* probabilities from **both** the Bond-Head (individual bond likelihoods) and the Count-Head (prior on number of breaks).
- This allows the stronger signal to win, meaning strong bond predictions can override a weak count prediction.

Overall Benchmarks: Outperforming G2G at Top-k > 1

Our model demonstrates competitive performance, especially at higher Top-k values.

Model / Setting	Top-1	Top-2	Top-3	Top-5	Top-10
Baseline	46.0	–	67.3	69.9	70.5
G2G ^[8] (reaction class known)	90.2	94.5	94.9	95.0	–
G2G ^[8] (reaction class unknown)	75.8	83.9	85.3	85.6	–
Our Best Model	64.9	–	91.5	95.8	98.4

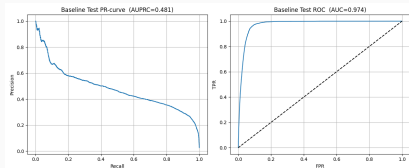
- Our model **outperforms G2G** (reaction class unknown) for $k \geq 3$.
- While G2G has a higher Top-1 accuracy (75.8% vs. 64.9%), our model overtakes at Top-3 and beyond, reaching **91.5% / 95.8% / 98.4%**.
- This indicates our model's strong ranking mechanism, placing variants of the correct center immediately after the first guess.

Baseline vs Best Model

Baseline Model

- A simple GNN without the multi-task count-head.
- Suffered from unstable training and lower performance.

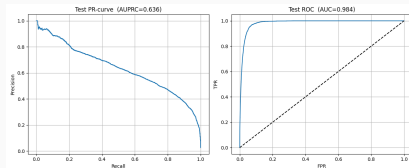
- **AUPRC: 0.481**



Final Model

- Incorporates a multi-task architecture.
- Uses graph-level features
- Uses a CosineAnnealingLR scheduler for stable training.
- Features a more robust inference strategy (**Beam Search**).

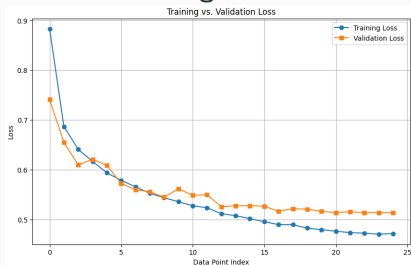
- **AUPRC: 0.636**



Training Losses

Both models show smooth convergence, indicating they are learning generalizable patterns and no overfitting.

Baseline Training



Final Model Training



Analyzing the Count-Head

- Its overall accuracy at predicting the correct number of breaks is **87.2%**.
- Confidently pinpoints < 1 reaction centers
- **Key Insight:** It consistently underestimates 2-bond reactions (64% of the time). This is where beam search provides the most benefit, by allowing two high-probability bonds to form a winning hypothesis.

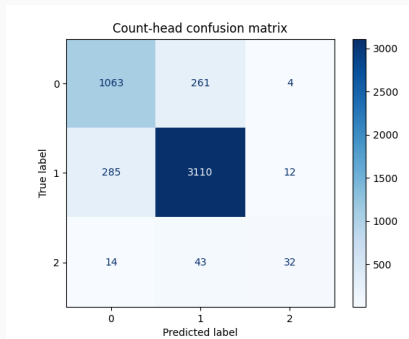


Figure 1: Confusion Matrix for the Count Head.

Qualitative Example Analysis

Success Cases

- The model's top prediction for the reaction center perfectly matches the ground truth.
- Often involves typical disconnections like C-N and C-O bonds adjacent to carbonyls or hetero-aromatic rings, reflecting learned textbook patterns.

Failure Cases

- *Over-confident false positive:*
The model predicts a plausible but incorrect bond, sometimes due to local structural features, overlooking broader context.
- *Underestimation of breaks*
- *Alternative valid disconnection:* The model identifies a chemically plausible disconnection different from the one recorded in the patent dataset.

Qualitative Examples on Test Set

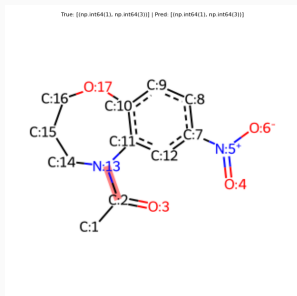


Figure 2: Correct C-N amide bond break.

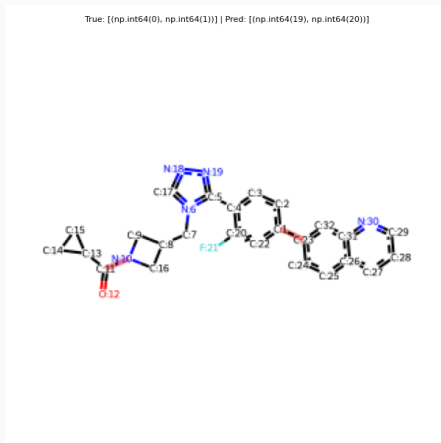


Figure 3: Incorrect bond, correct count (over-confident false positive).

Ablation Study: Impact of Beam Search

Beam search significantly boosts Top-k accuracy, especially for higher k values.

Setting	Top-1	Top-3	Top-5	Top-10
No Beam Search	47.0%	68.0%	70.2%	71.0%
With Beam Search	64.9%	91.5%	95.8%	98.4%

- The model's predictions are highly dependent on the Count-Head without beam search, leading to guaranteed failure if the count is wrong.
- Beam Search solves this allowing the model to explore multiple hypotheses, using the Count-Head's output as a "soft prior" rather than a rigid rule.

Ablation Study: Impact of Global Features

Global features are crucial

Setting	AUPRC	Top-1	Top-3	Top-5	Top-10
No Global (& No Multi-Task)	0.477	46.0%	64.%	68.8%	
No Multi-Task (with Global)	0.476	46.4%	65.0%	69.9%	
Global & Multi-Task (Our Best)	0.636	64.9%	91.5%	95.8%	

- We see a small drop between no multi with task and no global, climbing as k increases.
- And we see massive drops in all metrics compare to multi task
- Global features are the sole input to the Counting Head. Thus, we cannot ablate them separately

Ablation Study: Impact of Number of Hydrogens

The total number of hydrogens per atom contributes noticeably to performance.

Setting	AUPRC	Top-1	Top-3	Top-5	Top-10
No Hydrogens	0.600	61.7%	89.1%	94.4%	98.3%
With Hydrogens	0.636	64.9%	91.5%	95.8%	98.4%

- While the impact on Top-3, Top-5, and Top-10 accuracies is minimal, there's a noticeable drop when excluding hydrogen count.
- Approximately a $\sim 3\%$ **drop in Top-1 Accuracy** and a $\sim 3.5\%$ **drop in AUPRC** are observed.
- This shows that the local structural context, captured by hydrogen count, is important for the model's performance.

Achievements

- **Effective Multi-Task GNN:** Developed a GNN predicting bond breaks and count, leveraging diverse graph features.
- **Critical Beam Search:** Implemented beam search for robust inference, overcoming rigid count predictions and maximizing multi-task benefits.
- **Strong Performance:** Achieved **0.636 AUPRC** and high Top-k accuracies (e.g., **91.5% Top-3**, **98.4% Top-10**), outperforming a G2G baseline for $k \geq 3$.
- **Validated Ablations:** Confirmed the essential contributions of global features, multi-tasking, beam search, and hydrogen count.

Future Work

- **End-to-End Model:** Develop models that jointly identify reaction centers and generate reactant structures (e.g., semi-template or graph generation).
- **Advanced Features:** Incorporate richer chemical properties (e.g., bond enthalpy, partial charges) and 2D/3D stereometry.
- **Data Augmentation:** Synthetically duplicate multi-center reactions and introduce symmetry/ordering information.
- **Scale Up:** Train on larger datasets (e.g., MIT-USPTO) and explore transfer learning strategies.
- **Reaction Class Awareness:** Integrate reaction class labels to inform and boost prediction accuracy.